# Khang Nguyen

615-939-3645 | KhangN44345@gmail.com
github.com/KhangN65719 | linkedin.com/in/khangn445

## EDUCATION

**University of Illinois Chicago** | Chicago, IL
Bachelor of Arts in Computer Science | Expected May 2027
**Relevant coursework:** Data Structures & Algorithms, Software Design, Machine Organization, Programming Language Concepts

## TECHNICAL SKILLS

**Languages:** Java, Go (Golang), C++, Python, C, SQL
**Tools & technologies:** Git, GitHub, Docker, Railway, JavaFX, MySQL, SQLite, REST APIs, Maven
**Testing:** JUnit, Google Test (C++), unit testing

## PROJECTS

### Networked Checkers Application
*Java, JavaFX, TCP Sockets | Apr 2026 | Team of 2*

- Co-developed a Java client-server game with a multithreaded TCP socket server for player matchmaking, move relay, rematches, and disconnect handling.
- Implemented JavaFX gameplay with move validation, chained captures, king promotion, in-game chat, and a single-player computer opponent.
- Processed incoming TCP messages on a background thread and used Platform.runLater to update the JavaFX interface on its UI thread.

### Personal Shell
*C, POSIX System Calls*

- Built a command-line shell that launches programs with fork and execv, resolves executables through PATH, and waits for child processes with waitpid.
- Implemented command pipelines using pipe and dup2, with input, output, and standard-error redirection through file descriptors.
- Added command sequencing and conditional execution with && and || based on command exit status.

### GitHub Activity Bot
*Go, REST APIs, Docker, Railway | Jun 2026*

- Built a Go service that retrieves repository and commit activity from the GitHub REST API, decodes JSON responses, and delivers scheduled daily summaries through Gmail SMTP.
- Separated API access, report formatting, email delivery, and scheduling into dedicated packages; deployed to Railway with a multi-stage Docker build and Alpine runtime image.
- Grouped commit messages by repository and calculated daily totals for commits and repositories with activity.

### ScholarChain
*Scala 3, sbt, Docker | Aug 2026 - Present*

- Direct Codex agents on a distributed systems course project that aims to map coauthorship over time and answer questions with publication evidence.
- Reviewed and approved a multi-module Scala build with Docker packaging and a regression check that preserves the last successful build marker when tests fail.